

5G

Module 44

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Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala

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hello viewers I welcome you to the series of lectures on cyber security and information security today we will discuss 5G what is 5G fifth generation mobile networks or fifth generation Wireless Systems is abbreviated as 5G it is used to describe the upcoming fifth generation of mobile network technology it is the coming fifth generation Wireless Broadband technology which is based on the i e 802.11 AC standard 5G technology is going to be a new mobile revolution in Mobile Market through 5G technology now we can use Worldwide Cellular phones with the coming of cell phone alike to PDA now your whole office is on your fingertips or in your phone 5G technology has extraordinary data capabilities and has ability to tie together unrestricted call volumes and infinite data broadcast within latest

mobile operating system the following figure shows the speed of 3G 4G and 5G in the form of a dashboard meters what exactly is 5G the G in 5G stands for Generation the wireless phone technology technically started with 1G in the early 1990s and 1G expanded to 2G when people are able to send text messages between two cellular devices and this is made by only the companies after this the world moved towards 3G with this people were capable to make phone calls send text messages and browse the internet four generation enhanced many of the capabilities that were made possible with the third generation of Wireless people could browse the web send text messages and make phone calls and they could even download and upload large video files without any issues after 4G companies added LTE that

is long-term Evolution with 4G connectivity LTE became the fastest and most consistent variety of 4G compared to competing technologies like ymax the difference between ymax and LT is both Technologies achieved similar outcomes but it was important to create a standard for everyone to use LT makes the 4G technology faster 5G will provide better speeds and coverage than the 4G 5G works with a 5 GHz signal and is set to offer speeds of up to 1 gabit per second for tens of connections or tens of megabits per second for tens of thousands of connections 5G will build on the foundation created by 4G LTE it's going to allow people send texts make calls and browse the web as always but with the increase of speed 5D will make easy for people to download

and upload Ultra HD and 3D videos the Next Generation mobile networks Alliance defines the following requirements start of 5G standard should fulfill data rates of tens of megabits per second for tens of thousands of users data rates of 100 megabits per second for metropolitan areas 1 gbps per second simultaneously to many workers on the same office floor several hundreds of thousands of simultaneous connections for massive wireless sensor Network spectral efficiency is significantly enhanced compared to 4G coverage improved signaling efficiency enhanced 1 millisecond latency or very less we can say virtually zero latency smart communication latency reduced significantly compared to LTE so some of these features can be shown in this diagram so now we come to design of

5G mobile network architecture this is a model that proposes design of network architecture for 5G mobile systems is all IP based model for wireless and mobile networks interoperability the system consists of a user terminal which has a crucial role in a new architecture and a number of independent autonomous radio access Technologies within each of the terminals each of the radio access Technologies is seen as the IP Link to the outside internet World however there should be different radio interface for each radio access technology R in the mobile terminal for an example if we want to have access to four different RS we need to have four different access specific interfaces in the mobile

terminal and to have all of them active at the same time with aim to have this architecture to be functional is 5G really faster than 4G yes 5G is really much faster than 4G and their speeds will be significantly faster now times 4G LTE transfer speeds top out at about 1 gbps per second this means it takes about an hour to download a short HD movie in perfect conditions but the problem is that people rarely experience 4G's maximum download speed because of the signal can be disrupted by so many different things for example buildings microwaves other Wi-Fi signals and many more 5G will increase download speeds up to 10 gbps per second this means a full HD movie can be downloaded in seconds it will also reduce latency significantly we can say that it

provides the Pass 2 thousands of connected devices that are placed at our homes and offices concepts for 5G mobile networks the 5G terminals will have software defined radios and modulation schemes as well as new error control schemes that can be downloaded from the internet the development is seen towards the user terminals as a focus of the 5G mobile networks the terminals will have access to different Wireless Technologies at the same time and the terminal should be able to combine different flows from different Technologies the vertical handovers should be avoided because they are not feasible in case when there are many Technologies and many operators and service providers in 5G each network

will be responsible for handling user Mobility while the terminal will make the final Choice among different wireless mobile file Access Network providers for a given Service Such choice will be based on an open intelligent middleware in the mobile phone the system comprising of a main user terminal and then a number of independent and autonomous radio access

Technologies each of the radio Technologies is considered as the IP Link for the outside internet world how does it work there there are already huge group of major Global telecoms working to create worldwide standards around 5G as we already have 4G and 3G so experts think or expects it is compatible with them but till now most of the standards

are not solidified as in the most basic form cell phones are basically two way radios when we call someone then our phone converts our voice into an electrical signal it transmits that electrical signal to the nearest Cellar using radio wave The Cellar bounces the radio wave through a network of SARTS and then to our friends phone the same is happening when we send other forms of data like photos and video across the network as when a new mobile wireless technology comes along like 5G it is assigned a higher radio frequency 4G occupied the frequency bands up to 20 MHz but in the case of 5G it will likely to use the frequency band up to 6 GHz the reason new wireless Technologies occupy higher frequencies is because they typically aren't in use and move

information at a much faster speed the problem is that higher frequency signals don't travel as far as lower frequencies so multiple input and output in will probably be used to boost signals anywhere 5G is offered 5G is an opportunity to develop a more sustainable operator investment model if previous generations of mobile technology have taught us anything it is that as with each preceding generation 5G will unlock value in ways we cannot and will not anticipate services that were initially expected to have a negligible impact 10 minutes, 1 second became hugely popular example SMS while those expected to be next big thing that have slow to gain transaction example video calling with the development of 5G the

industry can expect a paradigm shift in the way that all the stakeholders in the mobile ecosystem play their role Regulators especially can have this opportunity to create healthier environments that stimulate continuing investment in Next Generation technology some of the business cases that have worked well for 3G and 4G Technologies may not be right ones for 5G by continuing work operators have the new paradigm difference between 5G and 4G 4G networks are being widely deployed around the world and although the technology continues to evolve the focus now switches to 5G the fifth generation of mobile networks dubbed 5G the Next Generation Network may still be several years away from being deployed but it is

the buzz World in telecommunications at the moment 5G is the next major phase of mobile telecommunication and follows the global adoption of 4G LTE that is long-term Evolution standards since 2012 a new telecommunication standard is introduced approximately every 10 years 2G was introduced in 1991 3G in 2001 and 4G in 2012 and 5G is expected to be made available in 2020 or 2022 the fifth generation of mobile technology does not have a defined universally a gr Pawn standard although a few new likely Technologies are emerging and whereas 4G connections offer average speeds of 2540 Mbps at home 5G is likely to offer speeds in excess of 500 Mbps many entities believe that 5G will be defined by its ability to offer 1 gbits per second mobile seconds to 10 simultaneously connections

and Chinese mobile Giant huai expects 5G to be at least 100 times faster than the fastest 4G LTE standard currently available a 4G plus speed test 5G could be up to 100 times faster 5G Technologies and frequencies 4G in the UK currently uses the LT standard with radio waves operating on the 800 mahz 18800 mahz and 2.6 GHz frequency bands the 1800 mahz band were repurposed by E and three to use with LD while as the other frequencies were made available to all networks in the 4G Spectrum auction back in 2012 who I believe that 5G will be based upon a combination of both new radio access Technologies are and existing Wireless Technologies such as LTE HSPA GSM and Wi-Fi whilst others such as Samsung and researchers at New

York University have discussed the concept of using millimeter wave frequencies millimeter wave frequencies lie between 3 to 300 mahz which is much higher than existing Network standards and offer the key benefit that are scarely used by other broadcast Technologies such as AM FM radio and TV which means it has the potential for greater speeds and the capability of more data to pass through it the millimet wave approach does not have some potential downfalls as these waves do not pass through solid objects very well which makes them difficult to sustain over long distances as a result any mm wave approach would use lots of smaller base stations whereas current mobile networks rely on a combination of large base

stations and smaller Anna in densely populated areas and it's expected that this approach will be used by all 5G standards to ensure maximum capacity and minimal latency as mentioned elsewhere there is no Universal 5G standard defined and when eventually launched we could see a variety of Standards operating under the 5G Banner similar to how both LTE and yaax and American LTE standard used by CDMA networks operate under the 4G Banner what are the benefits of 5G the potential benefits to 5G are vast for both businesses and consumers for the former the additional capacity and speed would allow for Greater Mobile working Wireless for consumers the speed should offer additional benefits within the ability of your

smartphone example a business using a 5G Network would mean employes can video conference from any location whilst for consumers 5G could allow you download a film to your smartphone in under a second aside from the greater speed another key benefit to 5G is the greater capacity current 4G contracts offer Promises of speeds up to 150 megabits per second on the go but leave must users with inclusive data allowances of less than 5G although most networks do offer plans with higher alliances at an extra charge due to the limited capacity 5G is expected to offer upwards of 1,000 times the capacity of 4G which would allow mobile networks to offer greater data allowances as part of their Mobile plans 4G speed test showing

latency of 49 milliseconds another widely agreed benefit of 5G regardless of standard adopted is reduced latency latency is defined as the amount of time taken for data transfer to take place example the buffering period when streaming a video or the loading period when opening a web page and as the number of users on 4G has increased so has the latency with

5G huai is targeting latency of less than 1 millisecond compared to latency in excess of 50 millisecond on 4G which should mean that your video starts playing as soon as you hit play the reduced latency will also have wide reaching benefits in video games automated equipment The Internet of Things Arena and augmented reality for the later especially reduced latency in 5G could deliver exceptional experiences without the delays currently faced when

using augmented reality whether it's the Oculus Rift gear VR or Microsoft Hol lens all of these can benefit from reduced latency in order to deliver the most realistic experience a likely additional benefit of 5G depending on the standard used is that it will allow current network coverage to expanded into hard to reach areas while West 4G is widely being rolled out across the UK there are areas where coverage remains minimal at best and with 5G networks will be able to reach further areas by deploying multiple smaller an which will allow signals to be emitted in multiple directions and even bounced off Solid Surfaces now what are the requirements of 5G 5G has better coverage area and high data rate at the edge of the cell it has

low battery consumption so battery life will be much longer availability of multiple data transfer path around 1 GBS per second data rate is easily possible security is more Energy Efficiency and spectral efficiency are good worldwide coverage about 90% reduction in network energy usage because of the advantages fifth generation wireless system is very much Essential now we come to the applications of 5G network network availability will be everywhere and will facilitate people to use their computer and such kind of mobile devices anywhere anytime it will make the world real Wi-Fi zone it allow people to use their mobile and laptops and other devices anywhere and at any time as network will be available

everywhere mobile IP address will be assigned to the devices depends on the connected Network and geographical position by using IP V6 addressing technology it will make unified Global standards for all by using 5G technology we can receive radio signals even at a higher altitude now we come to the disadvantages of 5G technology now as we know 5G technology is researched and conceptualized to solve all radio signal problem but because of some technical advancements of some of the geographical areas and some of the security reasons it has the following disadvantages research is still going on it and this is under process the speed claims this technology seems difficult to achieve as there are other devices and incompetent technology involved in other parts of the

world many of the old devices to be replaced as these are not competent to 5G which increases the cost developing infrastructure will also increase the cost security and private issues yet to be solved so now what is the future span of 5G the future enhancement of Nano core will be incredible as it combines with artificial intelligent one can able to control his intelligent robot using his mobile phone our mobile can automatically type the message what our brain thingss the Google hot Trends have rated the term 6G as the 17th most searched word in the search engines the iPod 6G comes in seven different colors and has a aluminium

body which makes the body strong to withstand constant daily usage so that's all for the 5G technology thank you viewers for watching this program